

BAKHTA, Russian Federation

WMO: 237760

Lat: 62.4629N

Lon: 89.0051E

Elev: 57

StdP: 100.64

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -44.0	(c) -41.0	(d) -45.9	(e) 0.0	(f) -42.7	(g) -43.5	(h) 0.1	(i) -40.2	(j) 10.0	(k) -17.4	(l) 7.4	(m) -19.2	(n) 0.3	(o) 320	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.1	(c) 28.3	(d) 18.6	(e) 26.5	(f) 17.8	(g) 24.7	(h) 17.3	(i) 20.2	(j) 25.2	(k) 19.2	(l) 24.0	(m) 18.2	(n) 22.7	(o) 2.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.5	22.4	17.4	12.6	21.2	16.4	11.8	20.5	58.6	24.6	55.1	24.2	51.9	22.6	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.1	(b) 6.8	(c) 6.0	(e) -46.5	(f) 31.4	(g) 2.0	(h) 1.3	(i) -47.9	(j) 32.4	(k) -49.1	(l) 33.2	(m) -50.2	(n) 33.9	(o) -51.6	(p) 34.9		
(a) 8.1	(b) 6.8	(c) 6.0	(e) -45.3	(f) 22.8	(g) 1.1	(h) 1.6	(i) -46.1	(j) 24.0	(k) -46.7	(l) 24.9	(m) -47.3	(n) 25.9	(o) -48.1	(p) 27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.7	-24.1	-19.8	-9.2	-1.8	4.2	15.8	18.7	14.4	7.0	-2.4	-16.5	-20.0		
	DBStd	15.97	9.78	8.51	7.40	5.31	4.87	5.04	3.63	3.77	4.30	5.38	8.71	10.13		
	HDD10.0	5267	1056	833	596	355	191	13	1	7	111	383	794	928		
	HDD18.3	7768	1314	1067	854	605	438	105	42	131	340	642	1044	1187		
	CDD10.0	630	0	0	0	0	12	186	269	143	21	0	0	0		
	CDD18.3	89	0	0	0	0	0	28	52	9	0	0	0	0		
	CDH23.3	689	0	0	0	0	4	252	382	52	0	0	0	0		
	CDH26.7	137	0	0	0	0	1	50	81	6	0	0	0	0		
(14) Wind	WSAvg	2.3	2.0	2.1	2.3	2.6	2.8	2.2	2.0	2.1	2.4	2.9	2.2	2.0		
(15) Precipitation	PrecAvg	710	46	37	40	42	55	66	67	74	70	81	71	60		
	PrecMax	880	88	64	74	69	114	145	112	118	107	131	125	93		
	PrecMin	481	3	17	10	11	25	20	11	22	15	20	44	29		
	PrecStd	108	19	12	16	15	21	25	25	24	23	21	20	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.3	-0.5	5.4	12.6	22.8	30.1	30.9	27.2	21.5	10.6	0.7	-1.4		
		MCWB	-4.0	-1.4	2.4	5.5	13.5	17.8	20.0	18.9	14.6	7.1	-0.2	-1.9		
	2%	DB	-6.2	-3.0	2.9	9.7	17.8	28.1	29.0	24.8	17.9	7.6	-0.8	-2.9		
		MCWB	-6.9	-3.7	0.4	3.9	11.3	17.9	19.5	17.6	13.3	5.8	-1.6	-3.5		
	5%	DB	-8.4	-6.3	1.5	6.7	14.2	26.2	27.2	22.7	15.1	5.9	-2.1	-4.3		
		MCWB	-9.0	-6.9	-0.3	2.5	9.3	17.2	18.4	16.8	12.0	4.6	-2.9	-4.8		
	10%	DB	-10.7	-8.7	0.0	4.8	11.2	24.3	25.6	20.6	13.4	4.3	-3.6	-6.8		
		MCWB	-11.2	-9.3	-1.4	1.4	6.7	16.7	17.9	16.1	10.8	2.9	-4.4	-7.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.1	-1.3	3.1	6.2	13.8	20.7	23.2	20.2	15.7	7.7	-0.1	-1.9		
		MCDB	-3.3	-0.7	4.8	11.3	19.8	25.1	28.6	25.2	19.1	9.1	0.5	-1.6		
	2%	WB	-6.9	-3.7	1.0	4.4	11.7	19.5	21.1	18.8	14.1	6.1	-1.4	-3.5		
		MCDB	-6.1	-2.8	2.3	8.5	17.2	24.9	26.7	22.6	17.0	7.5	-0.9	-3.0		
	5%	WB	-9.0	-6.8	0.0	3.1	9.6	18.5	19.8	17.7	12.4	4.7	-2.8	-4.9		
		MCDB	-8.5	-6.2	1.1	6.1	13.6	23.9	24.5	21.2	15.0	5.8	-2.1	-4.2		
	10%	WB	-11.1	-9.2	-1.3	1.9	7.5	17.4	18.8	16.6	11.0	3.1	-4.4	-7.2		
		MCDB	-10.5	-8.7	0.1	4.2	10.9	22.8	23.4	19.7	12.9	4.1	-3.6	-6.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	9.1	8.8	7.9	9.7	9.1	8.2	6.2	4.8	7.7	7.1		
		MCDBR	6.8	7.8	8.6	11.2	12.2	13.3	12.2	11.7	9.4	5.5	7.6	5.9		
		MCWBR	6.7	7.4	7.2	7.7	7.4	5.9	5.2	5.7	5.3	4.6	7.5	5.8		
		MCDBR	7.1	7.9	7.6	9.4	11.3	10.1	10.0	9.4	8.1	5.1	7.2	5.7		
	5% WB	MCWBR	7.0	7.5	6.7	7.0	7.4	5.6	5.2	5.4	5.2	4.5	7.3	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.294	0.334	0.357	0.374	0.387	0.444	0.410	0.352	0.312	0.243	0.233		
	taud		1.908	1.986	1.984	2.045	2.246	2.381	2.192	2.265	2.401	2.236	1.958	1.871		
	Ebn at Noon		355	615	738	813	839	836	766	760	746	612	366	187		
	Edh at Noon		40	78	114	133	119	107	126	108	78	61	35	22		
(44) All-Sky Solar Radiation	RadAvg		0.28	1.13	2.57	4.20	4.96	5.53	4.98	3.42	2.02	0.82	0.35	0.13		
	RadStd		0.04	0.09	0.23	0.51	0.55	0.52	0.51	0.39	0.28	0.13	0.07	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	+2.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page